

Steering Validation — the expression leg holds up under a real judge

MODEL Llama-3.3-70B-Instruct JUDGE Llama-3.3-70B (FP8) NULL 5 permutation seeds
JOB 8206479 2026-07-07

Re-run the negative control properly — **5 random-permutation seeds as a null** and a **stronger 70B judge** instead of the coarse 32B. Result: **10 of 11 candidate emotions are genuinely direction-specific** (real steering $\gg$ the random-vector null). The earlier pessimistic "4/11" was mostly a **weak-judge artifact** — a 32B judge misreading generic perturbation as emotion.

10_{/11}DIRECTION-SPECIFIC
(RIGOROUS)**0**_{/11}ARTIFACT (RANDOM
BEATS REAL)**5**RANDOM-SEED NULL
PER EMOTION**70B**

JUDGE (WAS 32B)

— ✓ RIGOROUS RESULT — REAL VS 5-SEED RANDOM NULL (LLAMA-70B JUDGE)

For each emotion: steer with its real mean_diff+gm vector vs **5 independent coordinate-permuted copies** (matched norm, random direction). **REAL lift** and each seed = max trait at coherence ≥ 70 , minus baseline. **direction-specific** $\Leftrightarrow$ REAL $>$ max(5 seeds) *and* $>$ null mean $+ 2\sigma$.

EMOTION	REAL	5 RANDOM SEEDS	NULL $M \pm \Sigma$	NULL MAX	VERDICT
thrilled	+74	10, 0, 10, 28, 24	14.5 ± 10.4	28	DIRECTION-SPECIFIC
exuberant	+69	13, 16, 27, 23, 16	18.9 ± 5.1	27	DIRECTION-SPECIFIC
euphoric	+62	25, 9, 15, 22, 20	18.4 ± 5.8	25	DIRECTION-SPECIFIC
playful	+54	5, 4, 14, 0, 1	4.8 ± 4.7	14	DIRECTION-SPECIFIC
stimulated	+51	30, 5, 21, 39, 10	21.0 ± 12.4	39	DIRECTION-SPECIFIC
sentimental	+25	8, 15, 10, 22, 12	13.4 ± 5.1	22	DIRECTION-SPECIFIC
amused	+22	12, 5, 10, 10, 5	8.4 ± 2.9	12	DIRECTION-SPECIFIC
alert	+20	6, 2, 2, 8, 2	4.2 ± 2.2	8	DIRECTION-SPECIFIC
aroused	+12	2, 2, 2, 0, 2	2.0 ± 1.0	2	DIRECTION-SPECIFIC
puzzled	+8	0, 0, 5, 2, 0	1.5 ± 2.0	5	DIRECTION-SPECIFIC
fulfilled	+50	51, 30, 36, 38, 30	37.0 ± 7.8	51	WEAK

VERDICT – THE LEG HOLDS

10/11 emotions clearly beat the random null; 0 are artifacts. Only **fulfilled** is borderline (real +50, but one of five random seeds also hit +51). The random directions score **low** (playful nulls ≈ 5 , aroused ≈ 2) while the real vectors drive trait up sharply (thrilled +74, exuberant +69) — the emotion vectors **do** causally, specifically steer expression.

— ! WHY THE FIRST PASS SAID "4" – THE JUDGE WAS THE CONFOUND

An earlier single-seed control with the **coarse Qwen-32B judge** found only 4/11 direction-specific and flagged amused / thrilled / aroused as "random wins." That was the judge, not the signal:

32B VS 70B

The 32B judge scored **generic perturbed text as "emotional"** — random vectors got trait lifts of +8...+17, swamping the real signal. The 70B judge **tells apart "specifically emotion-X" from "generically weird,"** so random nulls collapse to ≈ 2 –20 and the real vectors separate cleanly. Directly answers "does a better judge help?" — **yes, it flips the conclusion.** amused, thrilled, aroused all pass under the 70B judge.

— #2 BASELINE SATURATION — STILL REAL (ABOUT THE *COUNT* , NOT DIRECTION)

CONFIRMED

Neutral prompts dropped baseline from **~78 → mean 48.7**, so the original **"43/171 valid"** count was inflated by emotion-leaning eval prompts. This validation tests the **11 strong+moderate candidates** — of those, 10 are genuinely direction-specific. It does **not** re-audit all 171; the honest headline is "the strong/moderate emotions steer for real," not "43 do."

— #5 NORM-MATCHED STEERING — STILL BREAKS COHERENCE

NEGATIVE

--norm-match under the standard coefficient search collapsed coherence to **0–10**. Residual-norm-scaled injection is too strong; needs a separately-tuned gentle schedule, not a drop-in.

— BOTTOM LINE

VERDICT

- **Expression steering is real for the candidates we claimed it for** — 10/11 pass a matched-norm random-vector null under a competent judge. The vectors causally, direction-specifically induce emotion.
- **The pessimistic "~4" was a coarse-judge artifact**, not a null result. Judge quality was the dominant factor.
- **The "43/171" count is still inflated** (baseline saturation) — the right claim is about the strong/moderate emotions, not the raw count.
- **Preference ($r \approx 0.86$) and behavior legs unaffected; the headline dissociation result untouched.**

Caveats: the 70B judge is the **same model family** as the steered model (it scores text, not vectors, and real vs random are judged identically — but a fully independent GPT/Claude judge remains the gold standard). $n = 8$ prompts/condition; only the 11 strong+moderate candidates were tested, not all 171. **fulfilled** is genuinely borderline.

Method: 5 coordinate-permuted seeds (norm-preserving) as the random null; judge Llama-3.3-70B served FP8 on a dedicated h200 · job 8206479 (2× h200, real + 5 seeds, separate prompt_sets, --save-responses all) · `steering/run_steering_eval.py` · read-only analysis.